

Nathan Gardner Hattersley

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SUMMARY

Ph.D. candidate in Economics at the University of Texas at Austin with an M.S. in Statistics. Specialize in causal inference, pricing, demand estimation, and structural industrial organization using large-scale observational data. Experience spans antitrust litigation, policy research, machine learning, and production software engineering.

EDUCATION

- University of Texas at Austin** Austin, TX
Ph.D. Economics (expected 2026); M.A. Economics Aug 2020–Present
- University of Arizona** Tucson, AZ
M.S. Statistics; B.A. Mathematics and Computer Science Aug 2013–May 2018

SKILLS

- Programming:** Python, R, Julia, SQL, TypeScript, Rust, Java, C, Stata, MATLAB, Bash
- Economics & Statistics:** Causal inference, structural modeling, demand estimation, machine learning (XGBoost), nonparametrics, numerical optimization
- Tools:** Pandas, tidyverse, Postgres/PostGIS, spatial analysis, Docker, Git, HPC, Make/Snakemake, L^AT_EX
- Languages:** English (native); Spanish (proficient)

PROFESSIONAL EXPERIENCE

- Texas Office of the Attorney General** Austin, TX
Economics Fellow Jul 2024–Jun 2025
 - Analyzed antitrust cases involving algorithmic collusion, online advertising auctions, and digital markets using industrial organization methods.
 - Drafted economic analyses and reviewed expert rebuttals supporting litigation strategy.
 - Identified a security vulnerability affecting a key litigation matter, informing defensive legal strategy.
 - Presented technical findings to attorneys and translated quantitative analyses into accessible legal arguments.
- World Bank Group** Remote
Short-Term Consultant Feb 2021–Aug 2021
 - Built statistical indices measuring political participation in Afghanistan using large-scale survey data.
 - Developed algorithms for ordinal survey aggregation and PCA-based index construction.
 - Authored technical documentation and presented methods to economists and policy researchers.
 - Developed and led internal training on R, tidy data, functional programming, and reproducible workflows.
- JPMorgan Chase & Co.** Houston, TX
Software Engineer Jun 2018–Jun 2020
 - Developed and maintained production trading systems in Python and TypeScript supporting interest rate derivatives.
 - Designed regulatory workflows and trade models for new financial instruments.
 - Migrated legacy platform to microservice architecture using React, GraphQL, and WebSockets.
 - Collaborated with traders, product managers, and engineers to deliver production software.

SELECTED RESEARCH

- Dealer Consolidation and Vertical Relationships:** Causal analysis of automobile dealership consolidation using transaction-level data and geospatial analysis to estimate effects on prices and product availability.
- Dynamic Pricing:** Developed and estimated a dynamic structural model of manufacturer-dealer relationships. Combined structural estimation with machine learning (XGBoost) to recover underlying allocation and sales target policy of the manufacturer.
- Demand Elasticity and Curvature (Joint with Eugenio Miravete):** Nonparametric estimation of consumer demand to study the relationship between elasticity, curvature, pass-through, and market power, with comparisons against widely used demand systems.

ADDITIONAL EXPERIENCE

- Teaching:** Teaching assistant for Master’s Econometrics, Industrial Organization, and Causal Inference. Led weekly review sessions, designed programming assignments, and taught statistical computing.
- Research Computing:** Led workshops on high-performance computing for economics Ph.D. students and developed departmental guidance for parallel computing in Julia, Python, and MATLAB.